

WASSIM CHOUAYAKH

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EDUCATION

National Institute of Applied Science and Technology (INSAT)

Engineering Degree – Industrial IT and Automation (Expected 2028)

Tunis, Tunisia

Sep 2025 – Present

National Institute of Applied Science and Technology (INSAT)

Preparatory Cycle – Mathematics, Physics and Industrial IT

Tunis, Tunisia

Sep 2023 – Jun 2025

TECHNICAL SKILLS

Languages: C/C++ (bare-metal & HAL), Python, Assembly (ARM Cortex-M)

Microcontrollers: STM32 F1/F4 – HAL and register-level development, interrupts, DMA, real-time

Protocols: UART, I2C, SPI, CAN Bus – register-level driver implementation

Peripherals: GPIO, Timers, PWM (generation & input capture), ADC, EXTI

RTOS & OS: FreeRTOS (tasks, queues, semaphores), Linux/Unix, POSIX sockets, pthreads

Embedded Tools: STM32CubeIDE, STM32CubeMX, Keil uVision, ST-Link, Logic Analyzer, GDB

DevOps: Git, Docker, CMake, Make

PROJECTS

Virtual ECU Diagnostic Stack on Linux vCAN | C, SocketCAN, UDS, ISO 15765-2

2026

- Built a single-ECU diagnostic prototype supporting 5+ UDS services over Linux vCAN via SocketCAN and a C-based ISO-TP layer with under 10ms end-to-end response time
- Integrated Tester Present (0x3E) handling and negative response generation covering 10+ unsupported UDS service codes (NRC)
- Authored 12+ smoke tests covering the full ISO-TP request/response path and achieving 90%+ critical-path coverage

Real-Time Temperature Monitoring System via CAN Bus | STM32 F4, C, CAN, I2C, FreeRTOS

2025

- Architected and deployed a multi-node STM32 system with I2C sensor acquisition and bare-metal CAN Bus transmission to a central display node
- Orchestrated 4 concurrent FreeRTOS tasks (producer/consumer) for CAN Tx/Rx and ADC sampling, maintaining deterministic cycle times under 5ms
- Validated bus arbitration and timing with a Logic Analyzer; cut per-frame latency to under 1ms at 500kbps

STM32 Peripheral & Protocol Driver Library | STM32 F4, C, Bare-Metal, UART, I2C, PWM, DMA

2024

- Constructed 16+ bare-metal drivers (GPIO, Timers, PWM, ADC, UART, I2C) via direct register manipulation, bypassing any hardware abstraction layer
- Implemented PWM input capture and EXTI interrupt routines achieving sub-500ns latency across input frequencies up to 1MHz verified on a Logic Analyzer
- Wrote a full-duplex bare-metal UART driver enabling real-time bidirectional PC-to-STM32 serial communication and a register-level I2C driver for configuring and reading multiple external sensors

WORK EXPERIENCE

Information Systems Intern – PAMS

Sep 2026 – Present

UGFS North Africa – Tunis, Tunisia

- Contributing to the information system transformation program (PAMS), helping modernize core business processes
- Collaborating with business and IT teams on the deployment and adoption of the new system

AI Agent Developer – IaC Terraform Generator

Jul – Aug 2026

LOOYAS – Tunis, Tunisia

- LangGraph multi-agent pipeline (planner, contractor, coder, validator, security scanner, packager) generating Terraform for AWS, Azure and GCP from plain-text prompts
- Automated quality via terraform fmt/init/validate, TFLint, Checkov; deterministic HCL auto-fix + LLM refinement; stack: FastAPI, Celery, Redis, React, Docker Compose

Software & Validation Engineering Intern

Jul – Aug 2025

KPIT Technologies (Automotive Embedded) – Sfax, Tunisia

- Streamlined the AUTOSAR test-case compliance workflow by authoring a Python automation script, reducing manual review time by ~40% and cutting report processing from hours to minutes
- Delivered a cross-platform desktop tool (Flutter + Python backend) for internal validation workflows, adopted by 10+ embedded software developers

AI Engineering Intern

Jun – Jul 2025

Infosquare – Sfax, Tunisia

- Deployed a RAG-based chatbot (LLM + retrieval over 100+ files); ~30% better relevance, ~4h/week saved per analyst
- Automated ingestion (PDF, DOCX, TXT) via async FastAPI, 100+ documents per batch, zero manual intervention

LANGUAGES

French: Native **Arabic:** Native **English:** Fluent (professional) **Spanish:** Beginner